

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

1. Compare Star Topology and Bus Topology in terms of Cost, Reliability and Performance

Star Topology

In a star topology, every node is connected individually to a central device such as a switch or hub. Therefore, more cabling is required because each node needs a separate connection to the central device.

Bus Topology

In a bus topology, all devices share a single backbone cable. This reduces the amount of cabling and makes the network simple and economical to install.

Comparison

Criteria	Star Topology	Bus Topology
Cost	More expensive because it requires a central switch/hub and individual cables	Less expensive because it uses a single backbone cable
Reliability	Higher reliability; failure of one node does not affect other nodes	Lower reliability; failure of the main bus cable can affect the entire network
Performance	Better performance because communication is managed through the central device	Performance decreases as more devices are added because traffic increases on the shared medium
Installation	Requires more cabling and connections	Simple because of the single backbone
Fault effect	Individual node failure is usually isolated	Backbone failure can bring down the entire network

Analysis

Star topology has a higher initial cost, but this cost provides better reliability and performance. Since each device has its own connection to the central switch or hub, a problem with one node does not normally disturb the remaining nodes.

Bus topology is economical and easy to install, but all devices use the same communication medium. As more devices are connected, traffic increases and performance can degrade.

Example

A network using a central switch or hub with individual connections represents the star approach, while a network using one shared backbone cable represents the bus approach.

Conclusion

Star topology is better when reliability and performance are more important, whereas bus topology is preferable when low cost and simple

installation are the main requirements. Therefore, star topology provides better overall performance and reliability despite its higher cost.

2. Compare Mesh Topology and Star Topology in terms of Fault Tolerance

Mesh Topology

In a mesh topology, each node is connected to multiple other nodes. This creates multiple possible paths for data transmission.

Star Topology

In a star topology, all nodes are connected to a central hub or switch. Individual node failures can be isolated, but the central device becomes an important point of dependence.

Comparison

Criteria	Mesh Topology	Star Topology
Fault tolerance	Very high	Moderate
Alternative paths	Multiple paths are available	Normally depends on the central device
Individual link failure	Communication can continue through another path	Failure of an individual node does not normally affect others
Central device dependency	No single central hub is responsible for all connections	Central hub/switch is critical
Reliability	Very high	Good, but lower than mesh
Complexity	High	Lower than mesh

Analysis

The main advantage of mesh topology is redundancy. Because nodes have multiple connections, a failure in one link does not necessarily stop communication. Data can use an alternate path.

Star topology also provides reasonable fault tolerance because the failure of one node does not normally affect the other nodes. However, the central hub or switch is a critical component. If it fails, the entire network can stop functioning.

Therefore, mesh provides stronger protection against link failures, while star provides a simpler structure with moderate fault tolerance.

Example

If one connection in a mesh network fails, another available connection can be used for communication. In a star network, failure of the central switch or hub can affect the complete network.

Conclusion

Mesh topology is superior to star topology in terms of fault tolerance and reliability because of its multiple communication paths. However, star

topology is simpler and easier to manage, making it suitable when very high redundancy is not required.

3. Compare Bus Topology and Mesh Topology in terms of Installation Complexity

Bus Topology

Bus topology uses a single backbone cable to connect the devices. It requires fewer cables and fewer connections.

Mesh Topology

Mesh topology requires nodes to have multiple connections with other nodes. Therefore, the number of cables and ports increases significantly as more devices are added.

Comparison

Criteria	Bus Topology	Mesh Topology
Installation complexity	Low	Very high
Cabling	Single backbone cable	Large number of cables
Connections	Minimal	Many connections
Time required	Less time	More time
Cost of implementation	Low	High
Expansion	Relatively straightforward	Becomes increasingly complex
Maintenance	Simpler structure	Complex structure

Analysis

Bus topology is easier to install because all devices share a single backbone cable. The number of individual connections is minimal, making installation straightforward and suitable for small networks.

Mesh topology is considerably more complicated. Each node requires multiple connections, resulting in a large number of cables and ports. As the number of devices increases, the complexity grows significantly.

Although mesh topology provides excellent reliability and fault tolerance, its installation requires more time, equipment, and planning.

Example

For a small network requiring a simple setup, a single backbone-based bus structure is easier to implement. In a mesh structure, adding more nodes also increases the number of required connections, making the installation more complex.

Conclusion

Bus topology is much simpler, faster, and cheaper to install, while mesh topology is more complex and costly. Mesh is selected when reliability and multiple communication paths are more important than installation simplicity.

4. Compare Hybrid Topology and Star Topology in terms of Scalability

Hybrid Topology

Hybrid topology combines two or more different topologies, such as star, mesh, or bus. This combination provides flexibility when expanding a network.

Star Topology

Star topology connects all nodes to a central switch or hub. It can be expanded, but its expansion is limited by the capacity of the central device.

Comparison

Criteria	Hybrid Topology	Star Topology
Scalability	Very high	Good but limited
Expansion	New segments can be added flexibly	Additional hardware may be required
Flexibility	Very high	Moderate
Dependency	Can use different topology structures	Depends on central switch/hub capacity
Large network suitability	More suitable for flexible expansion	Suitable within the capacity of the central device
Complexity	Higher due to combined topologies	Lower

Analysis

Hybrid topology provides greater scalability because it allows different network structures to be combined according to organizational requirements. New network segments can be added without necessarily changing the existing structure.

Star topology is also scalable, but its expansion depends on the number of available ports on the central switch or hub. Once the device reaches its capacity, additional hardware is needed.

Therefore, hybrid topology provides more flexibility for organizations whose network requirements may change or grow.

Example

A hybrid network may combine star and bus structures or star and mesh structures. This allows different sections of an organization to use structures suitable for their specific requirements.

Conclusion

Hybrid topology provides greater scalability and flexibility than star topology because different topologies can be combined and new segments can be added according to organizational needs. Star topology remains simpler but is limited by the capacity of its central device.

5. Compare Star, Mesh, Bus and Hybrid Topologies in terms of Maintenance

Maintenance refers to how easily faults can be identified, isolated, repaired, and managed in a network.

Comparison

Criteria	Star	Mesh	Bus	Hybrid
Maintenance difficulty	Low	High	Moderate/High	Moderate
Fault identification	Easy	Difficult	Difficult	Moderate
Troubleshooting	Easier	Time-consuming	Difficult when backbone has a fault	Requires careful monitoring
Structure	Simple	Complex	Simple	Combination of structures
Reliability	Good	Very high	Lower	Depends on combined topologies
Administration	Easier	Requires more effort	Can be difficult during backbone faults	Requires skilled administration

Star Topology

Star topology is relatively easy to maintain. Since individual nodes have separate connections, a problem can usually be isolated to a particular node or cable. This makes troubleshooting faster.

Mesh Topology

Mesh topology is highly reliable but difficult to maintain because it contains many connections. Finding and troubleshooting problems can become time-consuming.

Bus Topology

Bus topology has a simple structure, but maintenance becomes difficult when the backbone cable develops a fault. Since the backbone is shared, identifying the exact fault can be challenging and the failure can affect the complete network.

Hybrid Topology

Hybrid topology has moderate maintenance complexity. It provides flexibility, but different combined topologies must be monitored carefully. Skilled administration may be required to manage the different sections.

Analysis

From a maintenance perspective, star topology provides the best balance between simplicity and reliability. Mesh provides excellent reliability but requires greater maintenance effort. Bus is simple in structure but backbone faults can be difficult to locate. Hybrid provides flexibility but requires proper monitoring because it combines multiple topology structures.

Conclusion

For easy maintenance, star topology is the most suitable among the four. Mesh is the most demanding because of its large number of connections. Bus has difficulty mainly when the backbone fails, while hybrid has moderate complexity because multiple topologies must be managed together.

Overall Conclusion

Each network topology has its own strengths and limitations:

- **Star: Best balance of performance, reliability, and easy maintenance.**
- **Mesh: Best for fault tolerance and high reliability.**
- **Bus: Best for simple and low-cost installation.**
- **Hybrid: Best for flexibility and scalability.**

Therefore, there is no single topology that is best for every situation. The appropriate topology should be selected based on the network's requirements for cost, reliability, performance, fault tolerance, installation, scalability, and maintenance.